Comparative US AQI Time Series Forecasting in California with Extreme-Event and Climate-Context Evaluation

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Abstract

California hourly Air Quality Index (AQI) dynamics exhibit strong short-term autocorrelation alongside sudden volatility during wildfire seasons. Standard annual averages mask model vulnerabilities during severe pollution spikes. We benchmark five tabular regressors and two baselines across 181,122 hourly records (2018–2025). A chronological split reserves the smoke-heavy 2020 season for validation, evaluating nowcasting and 24-hour forecasting on the 2025 holdout set. XGBoost achieves top $\boldsymbol{R^2}$ scores of 0.8711 (short-term) and 0.4648 (24-hour). Scenario slicing shows machine learning performance degrades sharply during upper-tail extremes, where persistence baselines match or beat complex models. Feature ablations show engineered vapor pressure deficit (VPD) and spatial lags offer marginal standalone accuracy gains. We operationalize these findings in a real-time early warning dashboard.

Keywords: US AQI forecasting, PM2.5, California, time-series regression, climate-context evaluation, vapor pressure deficit, spatial lag, XGBoost, extreme-event analysis

1 Introduction

Accurate air quality forecasts guide public health warnings and environmental regulation. In California, complex interactions between urban emissions, marine air masses, terrain, and wildfire smoke make hourly AQI prediction difficult. Models performing well under typical conditions often fail during severe pollution episodes. Random data splits also introduce severe temporal leakage due to autocorrelation.

This study benchmarks tabular machine learning models on a multi-city California dataset, addressing two main questions. First, how do gradient boosting and linear models perform when shifting from short-term nowcasting to a strict 24-hour horizon? Second, do these models maintain reliability when evaluated on isolated extreme events rather than full-year averages?

We present four key contributions. First, we construct a 181,122-record hourly panel combining EPA AQS observations and Open-Meteo weather fields [1, 2]. Second, we enforce a climate-aware temporal split reserving 2020 for wildfire-context validation and 2025 for final testing. Third, we test physically motivated features, including vapor pressure deficit (VPD) and cross-station spatial lags. Fourth, we combine global metrics with scenario stress tests and deploy an interactive alert dashboard.

2 Related Work

Air quality forecasting traditionally relied on statistical autoregressive models. While interpretable, linear methods struggle with complex weather interactions. Tabular machine learning provides flexible alternatives. Random Forest offers robust bagging-based ensemble predictions [3], whereas gradient-boosted decision trees—XGBoost, LightGBM, and CatBoost—capture non-linear environmental thresholds [4–6]. Spatial lag features from adjacent monitoring sites also enhance regional inference [7].

For wildfire PM2.5 estimation, Vu et al. [8] modeled the 2018 California Camp Fire using a weighted Random Forest. Merging AQS monitors, PurpleAir sensors, GOES-16 satellite data, and HRRR meteorology with SMOTE oversampling, they achieved an out-of-bag R^2 of 0.92 and a temporal cross-validation R^2 of 0.85. Their results show the value of combining ground and satellite data during extreme smoke events. We build on their work by incorporating engineered dryness metrics and spatial lags into a multi-year AQI forecasting panel.

Validation design remains critical. Random cross-validation leaks temporal information across adjacent hours, inflating performance [9, 10]. We prevent this by enforcing strict chronological splits and isolating the extreme 2020 wildfire season for validation.

3 Dataset and Exploratory Analysis

3.1 Study Area and Data Sources

The dataset covers Fresno, Los Angeles, and San Jose, pairing EPA AQS monitoring logs with Open-Meteo meteorological and AQI variables. The target variable, `target_aqi`, is a PM2.5-derived AQI proxy. Raw PM2.5 and PM10 proxy features are removed to prevent direct target leakage.

Preprocessing yields 181,122 hourly records with 21 base attributes prior to lag expansion. The target distribution displays strong right-skewness: mean AQI is 54.79, median is 54.54, the 95th percentile reaches 97.21, and the 99th percentile hits 151.84. This heavy tail requires scenario evaluation rather than relying solely on global error averages.

3.2 Exploratory Correlation Structure

Exploratory analysis confirms PM2.5 as the primary direct correlate of target AQI. Weather attributes provide secondary context; wind speed correlates negatively with AQI due to atmospheric dispersion. Figure 2 displays these relationships.

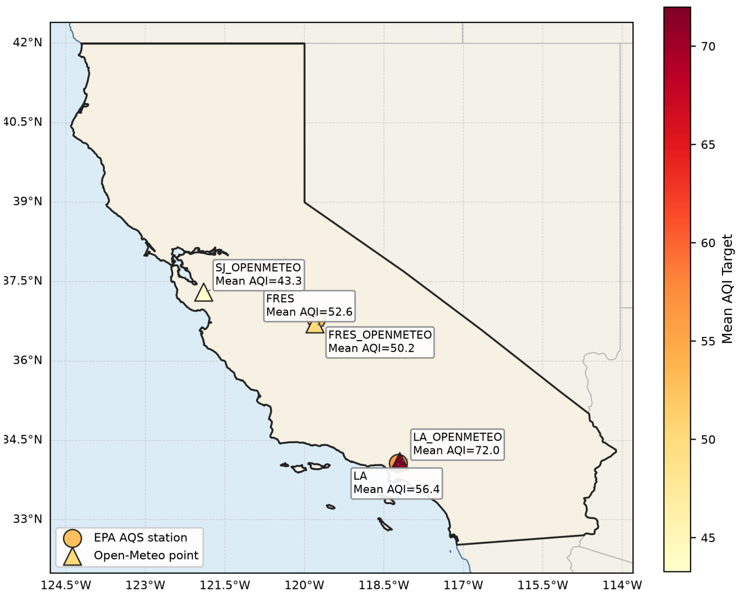


Fig. 1: Spatial coverage of the California AQI panel dataset.

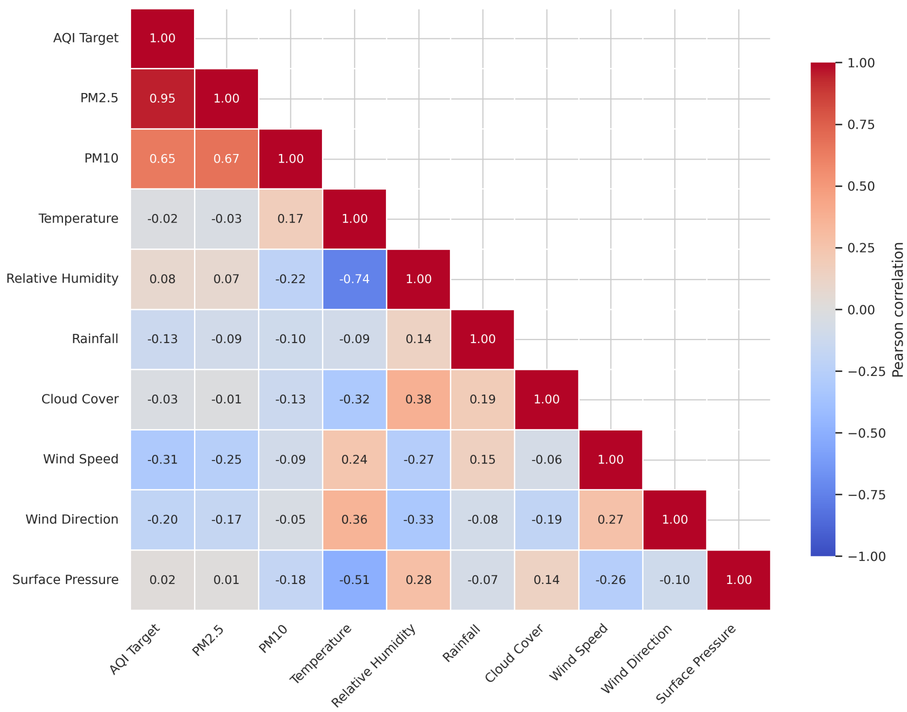


Fig. 2: Pearson correlation heatmap for selected AQI, pollutant, and meteorological variables in the model-ready California panel.

4 Feature Engineering

4.1 Engineered Dryness Feature: VPD

We compute Vapor Pressure Deficit (VPD) to quantify atmospheric moisture demand by comparing saturation and actual vapor pressure [11]. VPD indicates atmospheric dryness and plant water stress [12]. We evaluate VPD as a physical context variable rather than assuming prior accuracy gains.

Given temperature T (°C) and relative humidity RH (%), saturation vapor pressure is

$$e_s(T) = 0.6108 \exp\left(\frac{17.27T}{T + 237.3}\right). \tag{1}$$

Actual vapor pressure is

$$e_a(T, \mathrm{RH}) = \frac{\mathrm{RH}}{100} e_s(T). \tag{2}$$

The engineered VPD feature is

$$\mathrm{VPD} = \max\left(e_s(T) - e_a(T, \mathrm{RH}), 0\right). \tag{3}$$

Our exploratory code recomputes VPD from meteorological variables and generates a monthly diagnostic plot. This diagnostic verifies that the engineered dryness signal follows plausible seasonal structure before inclusion in the feature set.

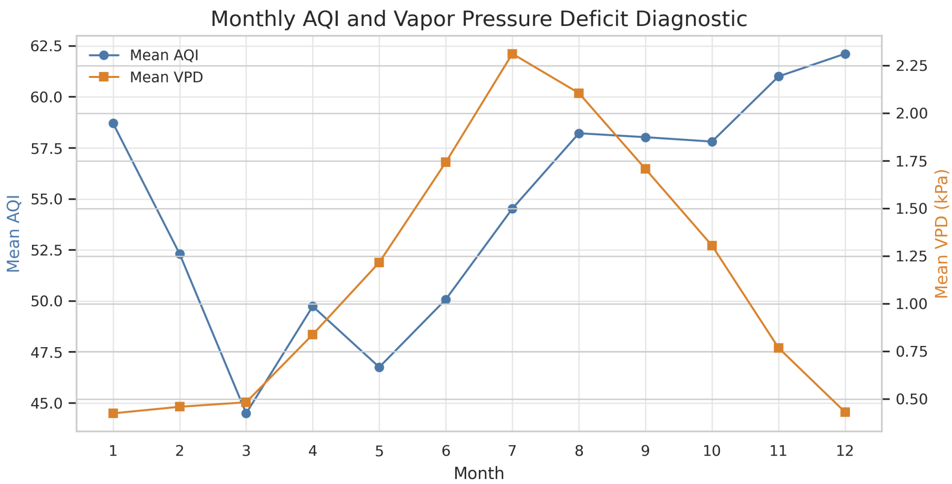


Fig. 3: Monthly diagnostic of recomputed VPD, inspecting seasonal dryness structure in the AQI panel.

4.2 Temporal and Spatial Lag Features

Temporal lag features encode recent and day-prior AQI states. For station s at target time t, a local lag feature is

$$\mathrm{Lag}_{s,k}(t) = y_{s,t-k}, \tag{4}$$

where k is the lag horizon in hours. Spatial lag features summarize delayed AQI information from other stations:

$$\mathrm{SpatialLag}_{s,k}(t) = \frac{1}{|\mathcal{N}(s)|} \sum_{j \in \mathcal{N}(s)} y_{j,t-k}. \tag{5}$$

These features provide transport-related context, and their predictive value is tested empirically in our ablation study.

Additional features include cyclic hour/month encodings, wind-vector decomposition, dry-heat interaction terms, dry-wind interaction terms, temperature-humidity ratios, and wind-speed-pressure ratios. Categorical station identifiers are retained for model families supporting categorical or one-hot encoded inputs.

5 Methodology

5.1 Pipeline Overview

Our workflow converts monitoring logs and weather inputs into a benchmark free of temporal leakage. Key stages include data harmonization, feature generation, chronological partitioning, model training, leaderboard scoring, scenario analysis, and ablation testing.

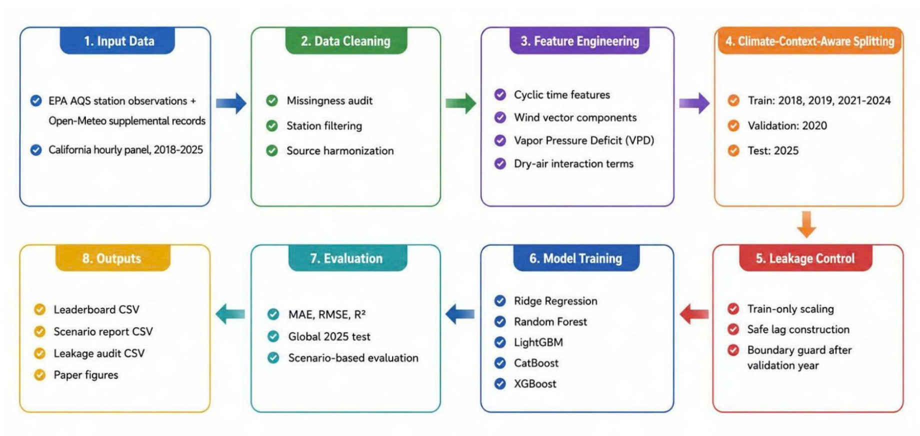


Fig. 4: AQI forecasting pipeline architecture.

5.2 Forecasting Tasks

For station s and target time t, the forecasting task is formulated as

$$\hat{y}_{s,t+h} = f(\mathbf{x}_{s,t}), \tag{6}$$

where $\hat{y}_{s,t+h}$ is predicted AQI at horizon h, and $\mathbf{x}_{s,t}$ contains weather features, calendar encodings, station IDs, VPD, lagged AQI, and allowed spatial context.

We evaluate two configurations:

1. **Short-term nowcasting**: Uses recent local lags (1–3 hours), leveraging strong temporal autocorrelation.
2. **24-hour forecasting**: Excludes local AQI from $t-1$ to $t-23$, forcing models to rely on weather, calendar cycles, VPD, spatial lags, and day-prior AQI.

5.3 Data Splitting Strategy

We enforce a chronological split to prevent temporal leakage. Random partitioning leaks lagged states and weather patterns across sets, artificially inflating performance. Table 1 outlines the split structure.

Table 1: Chronological data split used for model selection and evaluation.

Partition	Years	Role and rationale
Training	2018, 2019, 2021–2024	Model parameter fitting.
Validation	2020	Model selection under wildfire extremes.
Test	2025	Final independent performance reporting.

Isolating 2020 creates a high-smoke validation environment (August Complex fires). Unprecedented rainfall in 2022–2023 produced unusually clean air; training without extreme validation risks underestimating severe events. The 2025 test set (including the Palisades Fire) acts as the final holdout.

5.4 Chronological Boundary Leakage Guard

To prevent lag features from crossing split boundaries, we enforce a temporal buffer G_h:

$$\Delta t_{\text{boundary}} \geq G_h, \qquad G_h = \begin{cases} 4 \text{ hours}, & \text{short-term nowcasting}, \\ 215 \text{ hours}, & \text{long-term 24-hour forecasting}. \end{cases} \tag{7}$$

Audit checks verify that target proxies and future features are excluded.

5.5 Models and Metrics

We benchmark five regressors: Linear Ridge, Random Forest, LightGBM, CatBoost, and XGBoost. CatBoost processes categorical station IDs natively, while other models use train-set scaling and encoding.

Two operational baselines benchmark performance:

- **Persistence**: Projects the last available observation ($t-1$ for short-term, $t-24$ for long-term).
- **Climatology**: Predicts historical mean AQI per hour-of-day and month from the training set.

Evaluation metrics include Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|, \tag{8}$$

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}, \tag{9}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2}. \tag{10}$$

5.6 System Architecture and Deployment

We deployed trained pipelines into a real-time early warning web application. A FastAPI backend handles model inference and computes SHAP values for feature interpretability. An SQLite database caches Open-Meteo weather fields and EPA AQS data to minimize latency.

The Streamlit frontend displays interactive PyDeck heatmaps of AQI across Fresno, Los Angeles, and San Jose. It provides side-by-side timeline plots of actual versus predicted AQI. KPI summary cards trigger automated "Air Stagnation Alerts" during deteriorating weather or predicted AQI spikes.

6 Results and Discussion

6.1 Overall Leaderboard

The 2025 test leaderboard highlights clear differences between horizons. Near-term nowcasting yields high R^2 values across all models, whereas 24-hour forecasting drops sharply in accuracy.

6.2 Short-Term Autocorrelation Dominates Near-Term Nowcasting

Table 2 summarizes short-term nowcasting performance. Strong autocorrelation enables all models—including naive Persistence—to achieve $R^2 > 0.85$.

Table 2: Short-term autoregressive nowcasting results on the 2025 test year compared with the Reference Baseline Literature (RBL).

Model	MAE	RMSE	R^2
XGBoost	5.8321	9.4719	0.8711
LightGBM	6.0630	9.4897	0.8706
Linear Ridge	6.3248	9.6697	0.8657
CatBoost	6.2373	9.9989	0.8564
Persistence	6.5068	10.0480	0.8550
Random Forest	6.6052	10.1227	0.8528
Vu et al. [8] (RBL)*	–	–	0.8500
Climatology	19.3641	27.0662	−0.0525

Note: *RBL baseline by Vu et al. (2022) reported 10-fold temporal cross-validation $R^2 = 0.8500$ using satellite AOD and HRRR meteorology during the 2018 Camp Fire.

XGBoost achieves the top R^2 (0.8711) and lowest MAE. Persistence outperforms Random Forest and matches CatBoost, reflecting strong short-term temporal autocorrelation (0.927 correlation between target and 1-hour lag). Climatology fails ($R^2 = -0.0525$), missing transient spikes.

6.3 Horizon Degradation in 24-Hour Forecasting

Table 3 presents 24-hour horizon performance. Excluding local lags from $t-1$ to $t-23$ forces models to rely on weather, calendar patterns, VPD, and spatial lags.

Table 3: Long-term 24-hour forecasting results on the 2025 test year.

Model	MAE	RMSE	R^2
XGBoost	13.6366	19.3007	0.4648
LightGBM	13.5828	19.3349	0.4629
Linear Ridge	13.5179	19.3861	0.4601
Random Forest	13.9665	19.7806	0.4379
CatBoost	13.6376	20.2404	0.4114
Persistence	16.3495	23.1401	0.2307
Climatology	20.1250	28.0628	−0.1314

XGBoost leads ($R^2 = 0.4648$), followed closely by LightGBM and Linear Ridge. Persistence drops to $R^2 = 0.2307$, reflecting substantial 24-hour AQI shifts.

6.4 Upper-Tail Extreme Spikes Reveal Model Vulnerability

Global metrics mask operational vulnerabilities. Figure 5 illustrates AQI category shifts, highlighting high-pollution episodes.

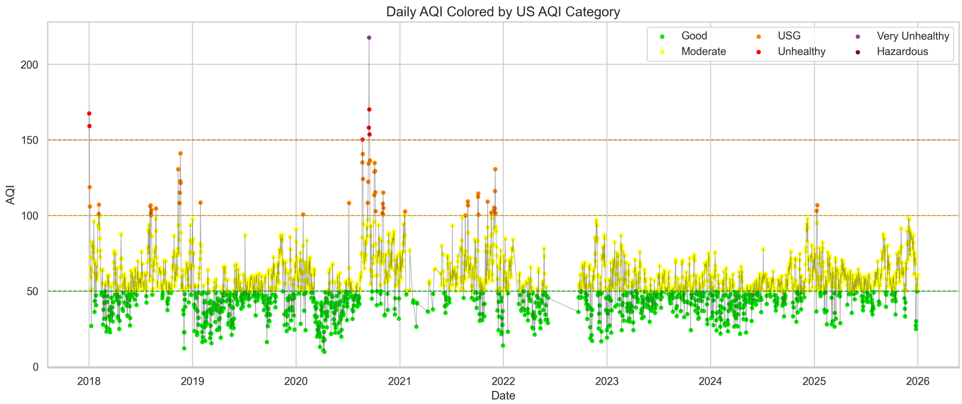


Fig. 5: Daily AQI observations colored by U.S. AQI health category.

Table 4 evaluates top models across specific scenario slices.

Table 4: Best-per-scenario results for extreme-event and climate-context evaluation on the 2025 test year.

Configuration	Scenario slice	Best model	N	MAE	RMSE	R^2
Short-term	Full 2025 test year	XGBoost	26293	5.8321	9.4719	0.8711
Short-term	Non-event baseline (AQI $\leq$ 100)	XGBoost	24806	5.1739	7.4436	0.8521
Short-term	January 2025 Winter Extreme Window	Linear Ridge	1795	9.2156	14.2454	0.8473
Short-term	Extreme AQI in 2025 (top 5%)	Persistence	1321	17.0277	23.0052	0.3004
Short-term	July–September 2025 wildfire season	Persistence	6624	5.9556	10.0932	0.8595
Long-term	Full 2025 test year	XGBoost	26224	13.6366	19.3007	0.4648
Long-term	Non-event baseline (AQI $\leq$ 100)	CatBoost	24756	11.2602	14.5048	0.4392
Long-term	January 2025 Winter Extreme Window	LightGBM	1726	19.4994	32.7758	0.2103
Long-term	Extreme AQI in 2025 (top 5%)	Persistence	1315	39.3547	51.7407	−2.3742
Long-term	July–September 2025 wildfire season	Random Forest	6624	13.3133	18.9501	0.5047

Three insights emerge. First, Persistence outperforms machine learning models on short-term top-5% AQI and Wildfire slices. Second, short-term top-5% R^2 drops to 0.3004. Third, 24-hour top-tail forecasts yield negative R^2 across all models (Persistence: −2.3742). Without smoke plume or fire perimeter inputs, weather and lag features cannot anticipate abrupt upper-tail spikes.

6.5 Prediction Diagnostics

Figure 6 plots XGBoost predictions against actual AQI. While central values match closely, high-AQI peaks display dispersion.

6.6 Comparison with Reference Baseline Literature (RBL)

We contextualize our empirical results against the reference baseline literature (RBL) by Vu et al. [8], who modeled PM2.5 levels during the 2018 California Camp Fire using a satellite-augmented Random Forest framework.

Vu et al. achieved a temporal cross-validation R^2 of 0.8500 and an out-of-bag R^2 of 0.9200 by integrating high-resolution GOES-16 satellite Aerosol Optical Depth

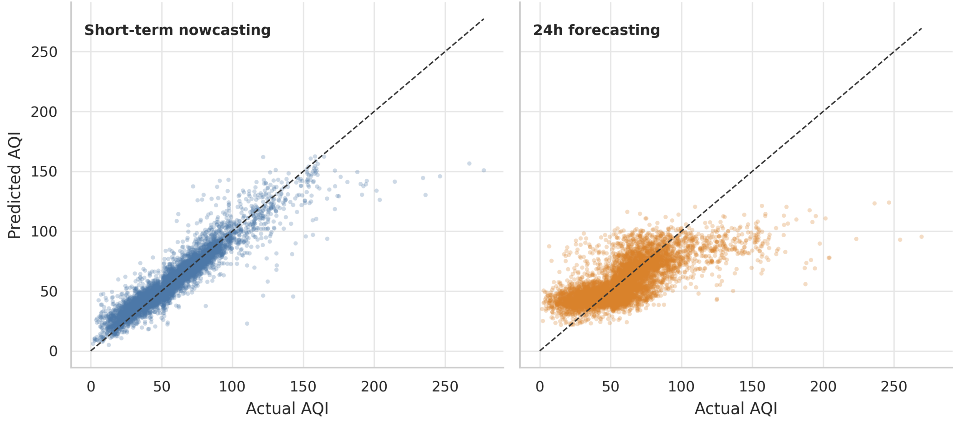


Fig. 6: Predicted versus actual AQI for XGBoost on the independent 2025 test year.

(AOD), HRRR meteorology, and SMOTE oversampling. In our benchmark, our top short-term nowcasting model (XGBoost) achieves a comparable test R^2 of 0.8711 (and Random Forest achieves $R^2 = 0.8528$), validating that tabular meteorological attributes paired with autoregressive AQI lags match near-term estimation performance.

Table 5: Comparative overview of our benchmark models against the Reference Baseline Literature (RBL) by Vu et al. [8].

Study / Model	Target / Scope	Key Feature Inputs	Validation Scheme	Test/CV R^2
Vu et al. [8] (RBL RF)	2018 Camp Fire	AQS + GOES-16 AOD + HRRR	Out-of-Bag (OOB)	0.9200
Vu et al. [8] (RBL RF)	2018 Camp Fire	AQS + GOES-16 AOD + HRRR	10-Fold Temporal CV	0.8500
Ours: XGBoost (Short-Term)	2025 Hourly AQI	AQS + Open-Meteo + Lags	2025 Test Set	**0.8711**
Ours: Random Forest (Short-Term)	2025 Hourly AQI	AQS + Open-Meteo + Lags	2025 Test Set	0.8528
Ours: XGBoost (24-Hour)	2025 Hourly AQI	Open-Meteo + Prior Lags	2025 Test Set	0.4648
Ours: Short-Term (Top 5% Slice)	2025 Upper Tail	AQS + Open-Meteo + Lags	Scenario Slice	0.3004
Ours: 24-Hour (Top 5% Slice)	2025 Upper Tail	Open-Meteo + Prior Lags	Scenario Slice	−2.3742

Two key operational insights emerge:

1. **Horizon Degradation**: While Vu et al. focused on short-term PM2.5 estimation during active fire windows, extending evaluation to a strict 24-hour horizon yields performance reduction ($R^2 = 0.4648$ for XGBoost).
2. **Extreme-Event Tail Vulnerability**: Under upper-tail extreme scenarios (top 5% AQI spikes), short-term accuracy drops to $R^2 = 0.3004$, and 24-hour predictions yield negative R^2. Tabular weather and lag features alone cannot anticipate sudden wildfire smoke transport without real-time satellite AOD and active fire perimeter inputs.

7 Ablation Study

Table 6 reports LightGBM ablation results removing VPD and spatial lags.

Table 6: LightGBM ablation study for VPD and spatial lag features on the 2025 test year.

Configuration	Feature set	Features	MAE	RMSE	R^2	ΔMAE	ΔR^2
Short-term	Full feature set	47	6.0630	9.4897	0.8706	0.0000	0.0000
Short-term	Without VPD	46	6.0901	9.5485	0.8690	0.0271	−0.0016
Short-term	Without spatial lags	32	6.1245	9.4834	0.8708	0.0615	0.0002
Long-term	Full feature set	49	13.5828	19.3349	0.4629	0.0000	0.0000
Long-term	Without VPD	48	13.6113	19.3162	0.4640	0.0285	0.0010
Long-term	Without spatial lags	34	13.5542	19.1877	0.4711	−0.0286	0.0081

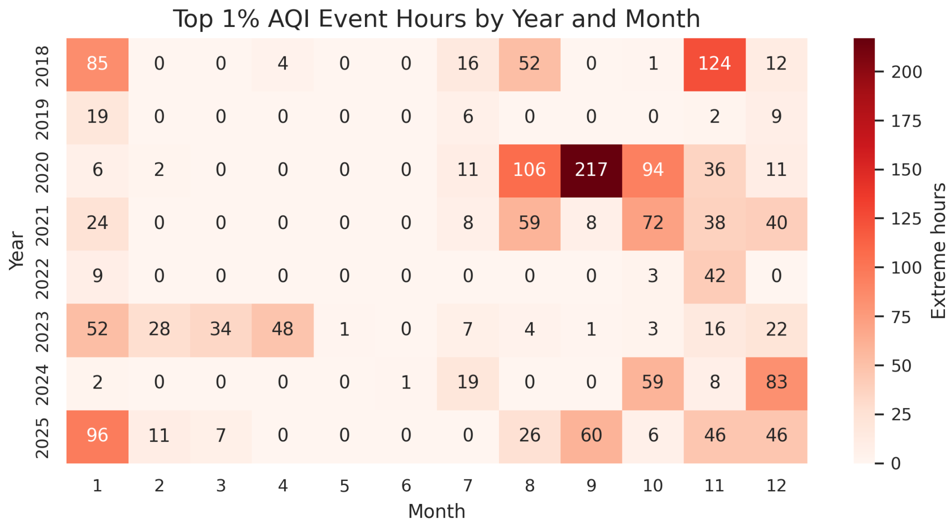
Removing VPD alters R^2 by < 0.003 across settings. VPD acts as an interpretable dryness descriptor rather than a standalone accuracy driver. Spatial lags show mixed impact: removing them slightly improves short-term R^2 while marginally raising long-term MAE.

8 Extreme-Event Analysis

We define extreme AQI using the 99th percentile threshold:

$$y_t > P_{99} = 151.84. \tag{11}$$

This isolates 1,802 extreme hourly records (0.99% of data). Year 2020 accounts for the largest share, with August–October containing 42.8% of extreme hours.



Year	1	2	3	4	5	6	7	8	9	10	11	12
2018	85	0	0	4	0	0	16	52	0	1	124	12
2019	19	0	0	0	0	0	6	0	0	0	2	9
2020	6	2	0	0	0	0	11	106	217	94	36	11
2021	24	0	0	0	0	0	8	59	8	72	38	40
2022	9	0	0	0	0	0	0	0	0	3	42	0
2023	52	28	34	48	1	0	7	4	1	3	16	22
2024	2	0	0	0	0	1	19	0	0	59	8	83
2025	96	11	7	0	0	0	0	26	60	6	46	46

Fig. 7: Year-month heatmap of extreme AQI records (99th percentile threshold).

Figure 7 shows strong temporal clustering during late summer and autumn, aligning with California wildfire seasons.

9 Threats to Validity

Four main limitations apply. First, `target_aqi` serves as a PM2.5 proxy rather than a full multi-pollutant AQI. Second, combining EPA AQS and Open-Meteo sources introduces potential data heterogeneity; 2025 relies predominantly on Open-Meteo records. Third, ablation reveals minimal standalone gain from VPD and spatial lags. Fourth, extreme-event clustering reflects correlation rather than causal fire attribution.

10 Conclusion and Future Work

We presented a California AQI forecasting benchmark across 181,122 hourly records (2018–2025). XGBoost achieved top performance in short-term nowcasting ($R^2 = 0.8711$) and 24-hour forecasting ($R^2 = 0.4648$). However, scenario analysis shows machine learning models struggle during upper-tail spikes without real-time smoke transport inputs, where naive persistence baselines remain competitive.

Future work will incorporate satellite Aerosol Optical Depth (AOD), fire perimeters, planetary boundary layer height, and wind vector transport. We also plan to evaluate Spatio-Temporal Graph Neural Networks (ST-GNNs) for regional modeling and connect model outputs to automated IoT public health alerts.

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Declarations

- **Funding**: Not applicable.

- **Conflict of interest**: The authors declare no competing interests.
- **Data availability**: The datasets generated and analysed during the current study are available from the EPA AQS and Open-Meteo public repositories.
- **Code availability**: Code and reproducibility materials are available upon reasonable request.
- **Author contribution**: All authors contributed to the study conception and design. Data collection, analysis, and implementation were performed by all authors. All authors read and approved the final manuscript.

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